Lecture Scribe: Inequalities and Tail Bounds

CSE400 – Fundamentals of Probability in Computing

1. Motivation: Tail Bounds

Key Idea

- The lecture begins with **tail bounds**, which are used to:
  - Estimate **probabilities of extreme events**
  - Understand **worst-case behavior**

Interpretation

- The “tail” refers to:
  - The probability that a random variable takes **very large or very small values**
- Motivation question (from case study):
  - *How to predict worst-case demands on a server?*

⇒ This motivates the need for **inequalities that bound probabilities without knowing full distributions**.

2. Markov’s Inequality

Definition (as per lecture structure)

- Introduced as the **first tail bound inequality**
- Applies to:
  - **Non-negative random variables**

Logical Flow (as implied in slides)

1. Start with a **non-negative random variable**
2. Relate:
   - Probability of large deviation
   - To its **expectation**

Proof

- The lecture includes:
  - **Formal definition**
  - **Proof (step-by-step in slides)**

(Exact derivation not present in provided content, so not reproduced here.)

Example

- An example is included in the lecture:
  - Demonstrates how to **apply the inequality to bound probabilities**

3. Chebyshev's Inequality

Definition

- Extends Markov's inequality
- Works with:
  - **Mean and variance**

Logical Flow

1. Start from:
   - Random variable with known **mean and variance**
2. Bound:
   - Probability of deviation from the mean

Proof

- The lecture explicitly includes:
  - **Definition**
  - **Proof**

Example

- Example demonstrates:
  - Bounding deviation using variance

4. Chernoff Bound and Poisson Tail

Definition

- A **stronger tail bound** compared to previous inequalities

Logical Flow

1. Used for:
    - Bounding probabilities of **sum of random variables**
2. Gives:
    - **Exponential decay bounds**

Proof

- Lecture includes:
    - Step-by-step proof

Example

- Shows:
    - Application for bounding rare events

5. Jensen's Inequality

Convex Functions

Definition

- Introduces concept of:
    - **Convex functions**

Logical Flow

1. Define convexity
2. Relate:
    - Function of expectation vs expectation of function

Jensen's Inequality

- Core idea:
    - Connects:
        * Expected value
        * Convex transformations

Proof

- Included in lecture:
  - Step-by-step reasoning

Example

- Demonstrates:
  - Application of inequality on convex functions

6. Case Study: System-Level Understanding

Problem

- Predict:
  - **Worst-case demand on a server**

Connection to Lecture

- Tail bounds help:
  - Estimate probability of **high load**
  - Provide **guarantees without exact distributions**

Summary Flow (Very Important for Exams)

1. **Motivation**
   - Need to bound extreme probabilities
2. **Markov's Inequality**
   - Uses expectation
3. **Chebyshev's Inequality**
   - Uses variance
4. **Chernoff Bound**
   - Strong exponential bounds
5. **Jensen's Inequality**
   - Uses convexity
6. **Application**
   - Real-world system (server demand)

Important Note

- The uploaded PDF **does not include actual derivations, formulas, or worked examples**—only the structure and topics.
- Therefore:
  - No additional derivations have been introduced
  - No assumptions beyond the document have been made